

Najmus Sadat Khan

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Executive Summary:

Najmus Sadat Khan is a water resources professional pursuing an MSc in Water Resources Development at IWMF, BUET. He focuses on numerical modeling, hydrological analysis, and flood risk assessment. His practical experience includes remote sensing and data-driven environmental modeling. He is highly skilled in using key industry software, including ArcGIS, HEC-RAS, HEC-HMS, and Delft-3D. He also has a strong background in assessing societal vulnerability to natural hazards like floods and cyclones. He works well in teams and communicates clearly.

Academic Qualification:

M.Sc. in Water Resources Development (Ongoing)

Institute of Water and Flood Management (IWMF), BUET | Session: April 2023 – July 2026(Expected)

B.Sc. in Civil & Water Resources Engineering

Military Institute of Science & Technology (MIST) | Duration: 2018 – 2022

- **Leadership Role:** President, MIST Environment Club (April 2021 – April 2022)

Experience:

Environment and Infrastructure Management Solution (EIMS) Limited (January 2025 – Present)

Associate Water Resources Engineer

- Leading the development of Anticipatory Action triggers for flash floods and drought to support vulnerable communities in hill tracts.
- Facilitating community-level consultations (FGD/KII) to assess local climate risks and coping capacities.
- Collaborating with the World Food Programme (WFP) to operationalize an impact-based forecasting framework for disaster response.

Red Cross Red Crescent Climate Centre (May 2024 – December 2024)

- Supported the development and review of Anticipatory Action (AA) protocols for cyclones and floods in Bangladesh.
- Assessed societal vulnerability to natural hazards by reviewing historical impact data and conducting exposure analysis.
- Coordinated with government agencies to improve the dissemination of flood and cyclone early warnings.

Environment and Infrastructure Management Solution (EIMS) Limited (May 2022 – April 2024)

- Conducted extensive field visits across 14 coastal districts to analyze climate risks and community preparedness for cyclones.
- Developed local-level thresholds for flash floods in the Haor region to enable timely evacuation and asset protection.
- Facilitated capacity-building workshops on Anticipatory Action for drought, training stakeholders to interpret climate data and implement early actions that reduce agricultural vulnerability.

Research & Publications:

Research in Progress

- Quantifying the Time of Emergence and Associated Uncertainties of Climate Change in Bangladesh using CMIP6 Projections.
M.Sc. Thesis, Institute of Water and Flood Management (IWFM), BUET.
Focus: Estimating emergence timing of climate change signals over Bangladesh using CMIP6 simulations and statistical detection methods.
- Imminent emergence of permanent marine heatwave states across global oceans.
Collaborative Research Project (in preparation for journal submission)
Authors: **Najmus Sadat Khan**, Hamza Ibn Mujib, Zahid Imtiaz, A.K.M. Saiful Islam, Thomas L. Frölicher, Robert Wilson, Juliane Wihgott, Mohammad Kamaruzzaman
Focus: Assessing trends and acceleration of marine heatwaves and anthropogenic forcing emergence using CMIP6 ocean projections and reanalysis data.
- Performance Assessment of CMIP6 Models in Reproducing Bangladesh's Rainfall and Temperature Using MCDM Ensembles and Individual GCMs Authors: **Najmus Sadat Khan**, Zahid Imtiaz, M. Shahjahan Mondal
- Developing Drought and Heat Wave Scenarios for Southwest Bangladesh Using Cmp6 Projections Authors: Zahid Imtiaz, **Najmus Sadat Khan**, M. Shahjahan Mondal

Conference Paper:

- **N.Sadat**, Zahid. S, MS Mondol, "PERFORMANCE OF CMIP6 GCMs IN SIMULATING RAINFALL PATTERNS OVER BANGLADESH" (**ICWFM-2025**)
- Zahid. S, **N.Sadat**, MS Mondol, "METEOROLOGICAL DROUGHT PREDICTION USING DIFFERENT FORECASTING MODELS" (**ICWFM-2025**)— **Selected for full paper submission as a book chapter** in *Climatic Resilience and Ecosystem Conservation in Bangladesh*, Springer (currently under review).
- S. Mehzabin, **N. Sadat**, M. Hassan, Saleh. AFM, "A STUDY ON CLIMATE INDUCED FLOOD INUNDATION OF BURIGANGA RIVER UNDER SHARED SOCIO-ECONOMIC PATHWAY SCENARIO" (**ICWFM-2023**)
- K. Nahid, S. Muni, **N. Sadat**, S. Ahmed, "STUDY ON RIVER BANK SHIFTING BY GIS AND REMOTE SENSING APPROACH OF JAMUNA BRAIDED BELT" (**ICACE-2020**)

Professional Development & Certifications

Green Skills for Climate Change Professionals | *University of Liberal Arts Bangladesh (ULAB), Center for Sustainable Development*

Professional Certificate | January – March 2025

- Received the highest mark (81.97%) among the participants.
- Developed essential green skills for climate adaptation and mitigation.
- Conducted field visits to climate-affected communities to assess vulnerabilities and adaptation strategies.
- Gained hands-on experience in project design, stakeholder engagement, and policy analysis.

Stormwater modeling with GeoSWMM | *Streams Tech Ltd, Dept of EWCE, MIST*

Departmental Course | September 2021 – January 2022

- Stormwater management modeling in an urban context.
- Scenario assessment using GeoSWMM.

Training & Capacity Building:

As Trainer | **Environment and Infrastructure Management Solution (EIMS) Ltd.**

November 2023 – December 2023

- Given three different trainings from November 2023 to December 2023 from EIMS to the **Bangladesh Water Development Board (BWDB)** officials. Training was on Data Processing, Model Development, Model Calibration, Model Validation, Forecasting, etc. on Groundwater modeling using MODFLOW, Salinity intrusion prediction and Drought Prediction modeling using machine learning.

January 2024

- Provided training on data collection for household survey using KOBO Toolbox to Bangladesh Red Crescent Society (BDRCS) volunteers for National EAP for Cold Wave Project.

April 2024

- Conducted a day-long technical training session titled **“Training on Drought Anticipatory Action Trigger Modelling”** at the **FAO Representation in Bangladesh, Banani, Dhaka**, under the project **“Increasing Capacities and Scale for Anticipatory Action Including through Social Protection Systems (ICSAASPS)”** (Project Code: OSRO/GLO/113/EC).
- Delivered hands-on exercises for Aman and Boro crop phases using Python-based drought trigger modeling and Combined Drought Index (CDI) analysis.

As Trainee | Capacity Building Training:

- March 2024: **Online course “Climate data analysis with python”**, UK Centre for Ecology & Hydrology, analyzing netcdf data plotting with Xarray (with certificate)
- August 2022: **Online course “Supervised Machine Learning: Regression and Classification”**, DeepLearning.AI & Stanford University (via Coursera), an introduction to machine learning in terms of regression analysis in python (with certificate)
- April 2022: **Online course “Exploratory Data Analysis with Seaborn”**, via Coursera Project Network, an overview of data analysis and data visualization with python (with certificate)
- January 2022: **Organized a workshop on “Programming Application in Water Resources and Environmental Management”** in MIST from MIST Environmental Club.

Technical Skills:

Engineering Application: ArcGIS, AutoCAD, HEC-HMS, HEC-RAS, SWMM, EPANET, Delft-3D, SPSS

Programming Language: Python, R, VBA

Reference:

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Bangladesh University of Engineering and Technology
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Syed Ahsanul Haque
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